

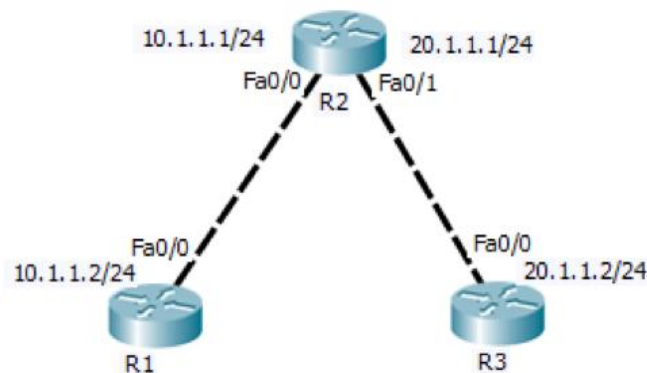
## DAY\_4

❖ **Introduction to OSPF Theory : How OSPF works: Areas, LSAs, and cost . Practical: Observe OSPF adjacency formation in Packet Tracer. Task: Write down steps to configure OSPF in a network.**

**1. Open Cisco Packet Tracer and create a simple network with three routers connected in a triangle topology using serial or Ethernet links.**

**ANS :**

➤ Network Topology



**2. Assign IP addresses to all router interfaces in your Packet Tracer network and verify connectivity using the ping command.**

**ANS:**

➤ Assign IP :

| Router | Interface       | IP Address | Subnet Mask   |
|--------|-----------------|------------|---------------|
| R1     | FastEthernet0/0 | 10.1.1.2   | 255.255.255.0 |
| R2     | FastEthernet0/0 | 10.1.1.1   | 255.255.255.0 |
| R2     | FastEthernet0/1 | 20.1.1.1   | 255.255.255.0 |
| R3     | FastEthernet0/0 | 20.1.1.2   | 255.255.255.0 |

➤ **PING Command :**

**From Router R1**

ping 10.1.1.1

**Result:** Success (100% success rate)

### 3. Configure OSPF on each router using the router ospf 1 command and the network command to advertise all directly connected networks.

ANS :

➤ **Router R1 Configuration**

```
enable
configure terminal

router ospf 1
network 10.1.1.0 0.0.0.255 area 0
```

➤ **Router R2 Configuration**

```
enable
configure terminal

router ospf 1
network 10.1.1.0 0.0.0.255 area 0
network 20.1.1.0 0.0.0.255 area 0
```

➤ **Router R3 Configuration**

```
enable
configure terminal

router ospf 1
network 20.1.1.0 0.0.0.255 area 0
```

➤ **Verify OSPF Configuration**

Run the following command on all three routers:

show ip protocols

Expected Output:

- Routing Protocol: **OSPF**
- Process ID: **1**
- Area: **0**
- Advertised Networks:
  - **R1:** 10.1.1.0/24
  - **R2:** 10.1.1.0/24, 20.1.1.0/24
  - **R3:** 20.1.1.0/24

**4. Observe and document the OSPF adjacency formation process by checking the output of the show ip ospf neighbor command on each router. Hint : Look for the FULL/DR or FULL/BDR state to confirm adjacency.**

**ANS:**

➤ **Command**

Run the following command on each router:  
show ip ospf neighbor

➤ **Router R1**

R1# show ip ospf neighbor

| Neighbor ID | Pri | State   | Dead Time | Address  | Interface       |
|-------------|-----|---------|-----------|----------|-----------------|
| 2.2.2.2     | 1   | FULL/DR | 00:00:39  | 10.1.1.1 | FastEthernet0/0 |

➤ **Router R2**

R2# show ip ospf neighbor

| Neighbor ID | Pri | State    | Dead Time | Address  | Interface       |
|-------------|-----|----------|-----------|----------|-----------------|
| 1.1.1.1     | 1   | FULL/BDR | 00:00:36  | 10.1.1.2 | FastEthernet0/0 |
| 3.3.3.3     | 1   | FULL/DR  | 00:00:34  | 20.1.1.2 | FastEthernet0/1 |

➤ **Router R3**

R3# show ip ospf neighbor

| Neighbor ID | Pri | State    | Dead Time | Address  | Interface       |
|-------------|-----|----------|-----------|----------|-----------------|
| 2.2.2.2     | 1   | FULL/BDR | 00:00:38  | 20.1.1.1 | FastEthernet0/0 |

➤ **Observation**

| Router | Neighbor | State           | Interface       |
|--------|----------|-----------------|-----------------|
| R1     | R2       | <b>FULL/DR</b>  | FastEthernet0/0 |
| R2     | R1       | <b>FULL/BDR</b> | FastEthernet0/0 |
| R2     | R3       | <b>FULL/DR</b>  | FastEthernet0/1 |
| R3     | R2       | <b>FULL/BDR</b> | FastEthernet0/0 |

**5. Change the OSPF cost on one interface using the ip ospf cost command and verify how it affects the OSPF routing table. Hint : Use show ip route ospf to see route changes after modifying the cost.**

**ANS :**

➤ **Step 1: Change the OSPF Cost**

On **Router R2**, increase the OSPF cost of the interface connected to **R3**.

R2> enable

R2# configure terminal

R2(config)# interface FastEthernet0/1

R2(config-if)# ip ospf cost 100

R2(config-if)# end

➤ **Step 2: Verify the OSPF Routing Table**

Run the following command on **Router R1**:

show ip route ospf

➤ **Sample Output**

R1# show ip route ospf

O 20.1.1.0/24 [110/101] via 10.1.1.1, FastEthernet0/0

Run the same command on **Router R3**:

R3# show ip route ospf

O 10.1.1.0/24 [110/101] via 20.1.1.1, FastEthernet0/0

➤ **Observation Table**

| Before Changing Cost                             | After Changing Cost                                                 |
|--------------------------------------------------|---------------------------------------------------------------------|
| OSPF cost was the default value (1).             | OSPF cost increased to <b>100</b> on FastEthernet0/1.               |
| Route metric was lower.                          | Route metric became higher ( <b>101</b> ).                          |
| Traffic used the only available path through R2. | Traffic still used the same path because no alternate route exists. |
| OSPF routing table showed a lower metric.        | OSPF routing table showed a higher metric.                          |